Project 1: Differentiated Demand for Cars and the VW–Fiat Merger

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1 Data, market definition, and price choice

The dataset used covers car purchases in five European countries (Belgium, France, Germany, Italy, and the UK) from 1970 to 1999. Markets are defined as country–year pairs, giving a total of 150 markets. For each, we observe the number of registrations for the 40 highest-selling cars.

The market size is defined to include agents who could potentially buy a vehicle: in line with the prevalent approach in the literature, I define the size as the number of households, which I constructed by dividing the population by the average household size in each country and year. Data on average household size are retrieved from the relevant national statistical sources. [1] Where data were missing, I extrapolated them linearly, assuming that the trend remained constant over time: the average household size across the five countries fell from 2.97 in 1970 to 2.38 in 1999.

This choice yields a remarkably large outside share, with a mean value of $s_{0t} = 0.923$ (Table 1). This is because cars are durable goods and only a fraction of households purchase a vehicle in any given year, and because the outside option also includes those who buy cars that do not rank in the top 40. The implications of this choice are discussed later in this exercise.

I use the variable `pr` taken in logarithm: since prices are recorded in nominal local currency over thirty years, taking the logarithm makes the price coefficient invariant to the currency unit, which is absorbed by market fixed effects. Moreover, the price includes VAT, as this is the price observed by the agent when deciding whether to buy a car. The firm's problem, however, is defined in terms of prices excluding VAT.

In the robustness check, I replicate the analysis using prices in levels.

[1] Italy, ISTAT; France, INSEE; Germany, Destatis, DZA, and DIW; the UK, ONS; and Belgium, Statbel.

2 Demand: model and identification

The utility that agent i derives from purchasing car j in market t, where $j = 0$ denotes the outside option, is given by:

$$u_{ijt} = x'_{jt}\beta - \alpha \ln p_{jt} + \mu_t + \lambda_{b(j)} + \kappa_{c(j)} + \xi_{jt} + \varepsilon_{ijt}, \qquad u_{i0t} = \varepsilon_{i0t}.$$

where

$$x'_{jt}\beta = \beta_1 \, \text{we}_{jt} + \beta_2 \, \text{li}_{jt} + \beta_3 \, \text{home}_{jt},$$

and x_{jt} contains the car's weight, fuel consumption, and a dummy indicating whether it is a domestic brand. I also include market (μ_t), brand ($\lambda_{b(j)}$), and class ($\kappa_{c(j)}$) fixed effects.

Under these assumptions, and assuming that the errors ε_{ijt} are independently and identically distributed (i.i.d.) Type I extreme value, market shares take the following logit form:

$$s_{jt} = \frac{\exp(\delta_{jt})}{1 + \sum_{k=1}^{J} \exp(\delta_{kt})}, \qquad s_{0t} = \frac{1}{1 + \sum_{k=1}^{J} \exp(\delta_{kt})},$$

where $\delta_{jt} = x'_{jt}\beta - \alpha \ln p_{jt} + \mu_t + \lambda_{b(j)} + \kappa_{c(j)} + \xi_{jt}$ denotes the mean utility of car j in market t.

Inverting the market-share equation gives:

$$\ln s_{jt} - \ln s_{0t} = x'_{jt}\beta - \alpha \ln p_{jt} + \mu_t + \lambda_{b(j)} + \kappa_{c(j)} + \xi_{jt}.$$

The model implies the following expressions for own- and cross-price elasticities:

$$\eta_{jkt} \equiv \frac{\partial s_{jt}}{\partial p_{kt}} \frac{p_{kt}}{s_{jt}} = \frac{\partial \ln s_{jt}}{\partial \ln p_{kt}}, \qquad \eta_{jjt} = -\alpha(1 - s_{jt}), \qquad \eta_{jkt} = \alpha s_{kt} \quad (j \neq k).$$

Since price enters utility in logarithmic form, a given percentage change in price has the same effect on mean utility, regardless of the car's initial price level. Under this specification, the marginal disutility of a one-euro price increase decreases as the car's price increases.

The logit specification implies the independence of irrelevant alternatives (IIA) property, meaning that cross-price elasticities are proportional to the market share of the car whose price changes. In the context of the merger, this means that customers who stop purchasing a model following a price increase are redistributed among the other alternatives in proportion to their market shares, regardless of the alternatives' characteristics. For example, if the price of a small car increases, consumers switch

to other small cars or SUVs according to their respective market shares, rather than according to how similar those cars are to the original model. The same logic applies to the outside option.

I assume consumers utility to be related to three observable car characteristics: weight, fuel consumption, and a dummy for domestic brands. Weight captures differences in car size and comfort, fuel consumption reflects running costs, and the domestic-brand dummy captures consumers' preference for domestic cars.

I also include market, brand, and class fixed effects to account for unobserved differences in average utility. Market fixed effects capture differences across country–year pairs, while brand fixed effects control for persistent differences in consumer preferences across brands. Class fixed effects account for differences across car categories.

The coefficient $\hat{\alpha}$ captures the effect of price on utility and is identified from price variation across cars, controlling for differences across brands, classes, and markets. However, unobserved car characteristics, ξ_{jt}, that are correlated with price may create an endogeneity problem.

To address this issue, I extend the model by estimating it using instrumental variables (IV) and two-stage least squares (2SLS), instrumenting price with the exchange rate of the country in which the car is produced. The relevance of the instrument is based on the following mechanism: if currency A appreciates against currency B, a car produced in country A becomes more expensive to sell in country B, leading the producer to charge a higher price in country B. This variation is not fully captured by the market fixed effects because it is related to the country of production rather than the destination market. It can therefore vary across the 40 cars observed in each market. I assume that the instrument is exogenous because it is driven by macroeconomic factors unrelated to car-specific characteristics.

Finally, the diversion ratio from product j to product k is defined as the proportion of customers lost by product j following a price increase who switch to product k. In the logit model, it is given by:

$$D_{j\to k,t} = \frac{\partial s_{kt}/\partial p_{jt}}{-\partial s_{jt}/\partial p_{jt}} = \frac{s_{kt}}{1-s_{jt}}, \qquad j \neq k.$$

An implication of the logit model is that diversion ratios depend only on market shares.

3 Results

The results are reported in Table 2. Both OLS and 2SLS estimates use heteroskedasticity-robust standard errors. The estimated price coefficient is -1.047 under OLS (standard error 0.092) and -4.334 under 2SLS (standard error 3.182). The first-stage F-statistic is 5.45, while the standard error of the 2SLS estimate is substantially larger than that of

the OLS estimate, indicating considerably lower precision.

Since individual car market shares are small in all markets, own-price elasticities are close to the estimated price coefficient in absolute value: approximately -1.045 under OLS and -4.326 under 2SLS. Thus, the 2SLS estimates imply substantially greater consumer price sensitivity than the OLS estimates.

Cross-price elasticities are also small. Their average values range from 0.0017 to 0.0021 under OLS and from 0.0068 to 0.0089 under 2SLS across countries. On average, a 1% increase in the price of one car is associated with an increase in the demand for another individual car of approximately 0.002% under OLS and 0.007–0.009% under 2SLS.

The diversion ratios reported in Table 3 are identical across the two models, as implied by the logit formula. Diversion between VW and Fiat is relatively limited, ranging from approximately 0.2% to 4.6%. By contrast, around 92–94% of customers lost by a car switch to the outside option. Diversion from VW to Fiat is highest in Italy, at approximately 4.6%, whereas diversion from Fiat to VW is highest in Germany, at approximately 2.6%. Since diversion ratios depend only on market shares, these patterns reflect Fiat's relatively large share in Italy and VW's relatively large share in Germany. However, diversion ratios alone are not sufficient to conclude that the merger will have its largest price effects in these two countries, as the effects also depend on demand elasticities and the firms' pricing incentives.

4 Merger simulation

Marginal costs are recovered from the multiproduct Bertrand–Nash pricing first-order conditions, using producer prices excluding VAT. Given the estimated demand, observed prices, and ownership matrix H, marginal costs are computed as:

$$\widehat{c} = p + (H \odot \nabla s(p))^{-1} s(p),$$

where $\nabla s(p)$ is the Jacobian matrix of market-share derivatives with respect to prices, and $\odot$ denotes elementwise multiplication.

Post-merger prices are obtained by solving the same pricing problem under the new ownership structure, holding marginal costs and consumer preferences constant. Moreover, the model assumes that technologies remain unchanged and the merger generates no cost savings.

Table 4 reports the estimated marginal costs and Lerner indices. The OLS estimates yield economically implausible results: 2.85% of estimated marginal costs are negative, while average Lerner indices range from approximately 0.96 to 0.98, with some exceeding one. By contrast, the 2SLS estimates imply consistently positive marginal costs and average Lerner indices of approximately 0.23–0.24. This difference is in line with the

lower price sensitivity estimated by OLS.

Table 5 reports the price changes implied by the merger under the two demand specifications. Under 2SLS, the largest price increases occur in Italy for VW (1.37%) and in Germany for Fiat (1.04%). These results are consistent with the diversion patterns discussed above and the relative importance of the two brands in their respective markets: Fiat has a stronger presence in Italy, while VW has a stronger presence in Germany.

The estimated price effects on competing firms are negligible, suggesting that the merger's indirect effects on competitors are limited.

5 Robustness check

As a robustness check, I re-estimate the 2SLS specification using price in levels rather than in logs. Specifically, I replace `pr` with `princ`, defined as the car price divided by nominal GDP per capita in the destination country. This provides an a-dimensional measure of price relative to per-capita income, facilitating comparisons across markets.

When price enters in levels, the own- and cross-price elasticities are given by

$$\eta_{jjt} = -\alpha \widetilde{p}_{jt}(1 - s_{jt}), \qquad \eta_{jkt} = \alpha \widetilde{p}_{kt} s_{kt}, \quad j \neq k,$$

where $\widetilde{p}_{jt} = \texttt{princ}_{jt}$ and $\alpha > 0$ is the negative of the estimated price coefficient.

The results are reported in Tables 2. The estimated coefficient on `princ` is -4.678 (standard error 3.125), with a first-stage F-statistic of 6.93. Mean own-price elasticities vary across countries, ranging from -2.827 in Germany to -4.453 in the UK. They also differ across car classes, increasing in absolute magnitude from 2.421 for class 1 to 6.737 for class 5, as expected. The implied merger price changes are reported in Table 5. The largest increases are estimated for VW in Italy (1.49%) and Fiat in Germany (1.28%).

6 Discussion

The two main specifications produce substantially different demand estimates. The OLS specification yields economically implausible results, as some estimated marginal costs are negative. This suggests that the model may not adequately capture firms' pricing behaviour, most likely because prices are correlated with unobserved product characteristics. The instrumental-variable approach is intended to address this endogeneity problem, and the estimated marginal costs are all positive under 2SLS. This difference is also reflected in the merger simulation results.

The first-stage F-statistic raises concerns about instrument strength, suggesting that the producer-country exchange rate explains only a limited share of the variation in prices. Possible explanations include the absorption of price variation by brand fixed

effects or limited exchange-rate pass-through by firms. Moreover, although the 2SLS point estimate can be used for the analysis, its standard error is large, resulting in a wide confidence interval. Consequently, estimates of α below one cannot be ruled out statistically.

The difference between the OLS and 2SLS merger results is driven primarily by the estimated value of $\hat{\alpha}$, since the substitution patterns and diversion ratios remain unchanged across the two specifications. The implied markups also differ substantially. Under OLS, markups are considerably higher, implying stronger incentives to raise prices following the merger. By contrast, the lower Lerner indices estimated under 2SLS indicate lower markups and weaker incentives to increase prices.

The robustness check suggests that the results are broadly similar when price is measured in levels rather than in logs, although the estimated merger price effects are slightly larger. Using price in levels changes the distribution of estimated markups across products. One possible explanation for the larger merger effects is that VW and Fiat produce cars concentrated in lower classes, where demand may be less price-sensitive. If this is the case, the incentives to retain customers following the merger may be stronger. Nevertheless, the overall conclusion remains unchanged, as the estimated price effects are comparable across specifications.

Overall, the largest price increases occur in Italy for VW and in Germany for Fiat, and this pattern is robust across specifications. These results appear consistent with the observed market shares. However, the magnitude of the price increases depends on the estimated price sensitivity, $\hat{\alpha}$, and is therefore subject to greater uncertainty.

Selecting only the 40 best-selling cars may affect both diversion ratios and the estimate of α. If consumers switch to VW or Fiat models outside the sample, the model treats them as choosing the outside option. This may underestimate how many customers switch between the two firms and, therefore, the merger's price effects. On the other hand, the sample may include cars with higher unobserved quality, which could make $\hat{\alpha}$ too close to zero. This would make consumers appear less sensitive to price changes and lead to higher estimated markups and possibly larger merger price effects.

Due to time constraints, I could not extend the analysis to estimate changes in consumer surplus or the break-even cost savings required to offset the merger's effects, as requested in the project. These extensions would provide a more complete assessment of the merger's welfare implications. Given the relatively small price effects estimated under 2SLS, the associated welfare effects may also be limited, although this cannot be established without calculating them.

7 Theoretical questions

2.1 A possible way to represent the data is through Bertrand competition with private information about costs. If firms have privately observed costs, they may set different prices and have different probabilities of winning consumers. To allow multiple firms to participate in the competition, I extend the model to include multiple draws from the cost distribution. However, in the standard model, where costs are independently drawn from the same distribution, firms are ex ante symmetric and therefore have the same probability of winning.

2.2 In the Perloff Salop model, the randomness $\varepsilon_{ij} \sim F$ captures the utility that consumer i derives from good j and is assumed to be i.i.d. across consumers and products. In the Logit model, utility is given by $\delta_{jt} + \varepsilon_{ijt}$, the sum of a mean utility common to all consumers and an individual-specific random shock. If we assume that ε in the Perloff–Salop model follows a Type I Extreme Value (T1EV) distribution, the two models become similar, except that the Logit model includes an additional term capturing the quality of each car, based on its characteristics, which is common to all consumers. Put differently, the Perloff–Salop model accounts for individual differences in preferences, whereas the Logit model also includes a product-specific mean utility. The two models also differ in their treatment of firms: the Perloff–Salop model does not allow for multiproduct firms and therefore cannot capture the effects of a merger.

2.3 -

2.4 In the Perloff–Salop model, the markup is a function of the number of firms in the market and its distribution. For example, in the case of the uniformly distributed shocks, we have seen that $p - c = \frac{1}{n}$, which implies that markups moves toward zero when the number of firms increases. When considering a normal or a Gumbel distribution instead, the result depends on the density around the marginal consumer, as the firms optimum problem shows, or in other words how distant consumers are between their best and second best option: with the Normal distribution, markups eventually converge to zero, while with the Gumbel, as n increases, the markup converges to a positive constant.

For the project, this implies that each firm has a constant markup, even if we were to include new car types in the market: this is because they would match some unusual taste in consumers.

Output

Table 1: Market shares by country (household market size)

Country	Mean inside share s_{jt}	Mean outside share s_{0t}
Belgium	0.0019	0.923
France	0.0020	0.918
Germany	0.0020	0.920
Italy	0.0020	0.919
UK	0.0016	0.937
All markets	0.0019	0.923

Notes: M_t = population / average household size. 150 markets, 40 cars per market.

Table 2: Demand estimates (dependent variable: $\ln s_{jt} - \ln s_{0t}$)

	(1) OLS	(2) 2SLS	(3) Robustness
ln `pr` (log price incl. VAT)	-1.047	-4.334	
	(0.092)	(3.182)	
`princ` (price / income per capita)			-4.678
			(3.125)
Weight (kg)	0.0005	0.0038	0.0030
	(0.0001)	(0.0031)	(0.0024)
Fuel consumption (l/100 km)	-0.052	-0.011	0.020
	(0.011)	(0.041)	(0.057)
Home	1.249	1.263	1.189
	(0.023)	(0.029)	(0.046)
Brand, market, class FE	Yes	Yes	Yes
Instrument	—	ln(`avexr`)	ln(`avexr`)
First-stage F	—	5.45	6.93
95% CI price coefficient	$[-1.23, -0.87]$	$[-10.57,\ 1.90]$	$[-10.80,\ 1.45]$
R^2	0.539	0.416	0.381
Observations	6,000	6,000	6,000

Notes: Heteroskedasticity-robust standard errors in parentheses.

Table 3: Group shares and diversion ratios between VW and Fiat, mean by country

Country	S_{VW}	S_{Fiat}	$D_{\text{VW}\to\text{Fiat}}$	$D_{\text{Fiat}\to\text{VW}}$	$D_{\text{VW}\to 0}$	$D^{\text{in}}_{\text{VW}\to\text{Fiat}}$	$D^{\text{in}}_{\text{Fiat}\to\text{VW}}$
Belgium	0.0126	0.0048	0.005	0.013	0.934	0.074	0.171
France	0.0051	0.0070	0.007	0.005	0.923	0.091	0.069
Germany	0.0257	0.0037	0.004	0.026	0.944	0.067	0.335
Italy	0.0070	0.0454	0.046	0.007	0.926	0.613	0.181
UK	0.0033	0.0020	0.002	0.003	0.940	0.034	0.054

Notes: $D_{A\to B} = S_B/(1 - S_A)$; identical across all models. D^{in} excludes the outside option (share of switchers to another inside car who go to the rival group).

Table 4: Implied Lerner indices, mean by country

Country	OLS	2SLS	2SLS, robustness
Belgium	0.964	0.233	0.372
France	0.969	0.234	0.363
Germany	0.965	0.233	0.395
Italy	0.971	0.235	0.286
UK	0.962	0.232	0.250
Max. Lerner (all markets)	1.093	0.264	0.827
Share of $\hat{c} \leq 0$	2.85%	0%	0%

Notes: Marginal costs from the multiproduct Bertrand–Nash FOC, using producer prices net of VAT.

Table 5: Merger simulation: mean percentage price change by country and group

	OLS			2SLS			2SLS, robustness		
Country	VW	Fiat	Others	VW	Fiat	Others	VW	Fiat	Others
Belgium	7.00	28.93	0.03	0.10	0.40	0.00	0.13	0.44	0.00
France	10.99	11.49	0.04	0.15	0.16	0.00	0.22	0.19	0.00
Germany	6.19	74.06	0.04	0.09	1.04	0.00	0.14	1.28	0.00
Italy	98.02	16.35	0.10	1.37	0.22	0.00	1.49	0.23	0.00
UK	3.49	8.69	0.01	0.05	0.12	0.00	0.05	0.10	0.00

Notes: Post-merger Bertrand–Nash prices computed with PyBLP, holding marginal costs fixed (no efficiencies).